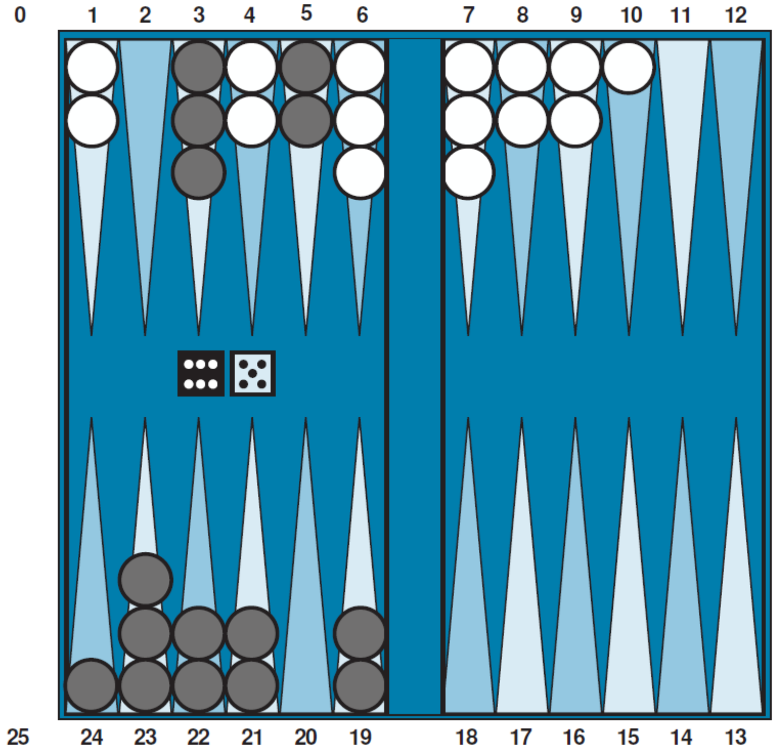
Stochastic Games

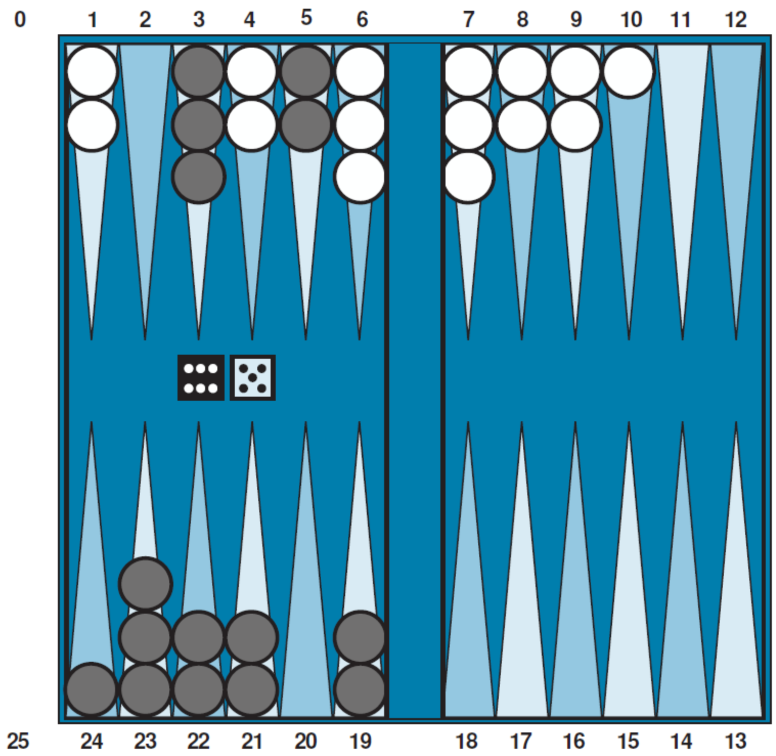
Some games (e.g., backgammon) have randomness due to throwing of dice. They combine both luck and skill.



* Figures/images are from the textbook site (or by the instructor).

Stochastic Games

Some games (e.g., backgammon) have randomness due to throwing of dice. They combine both luck and skill.

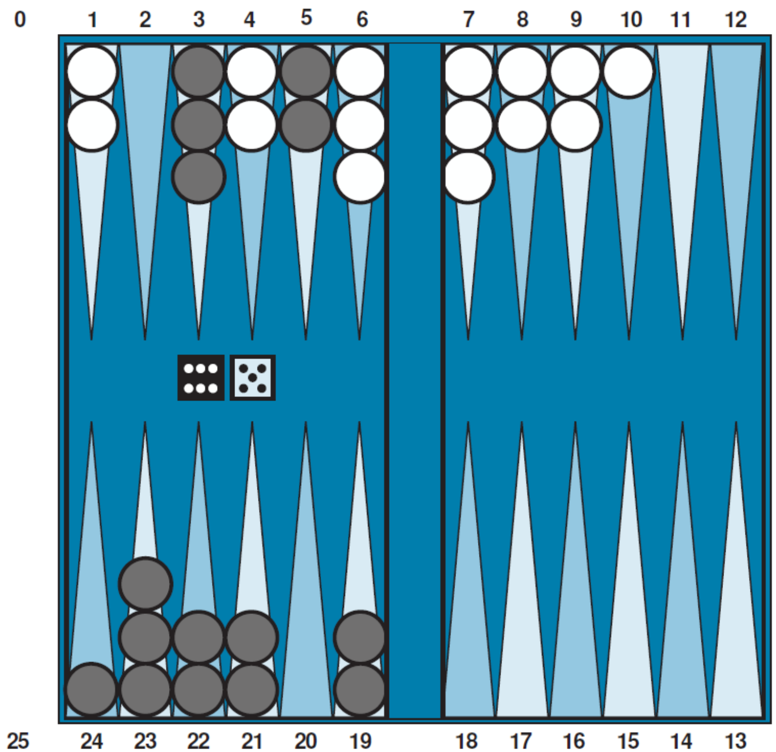


- Include *chance nodes* in addition to MAX and MIN nodes.

* Figures/images are from the textbook site (or by the instructor).

Stochastic Games

Some games (e.g., backgammon) have randomness due to throwing of dice. They combine both luck and skill.



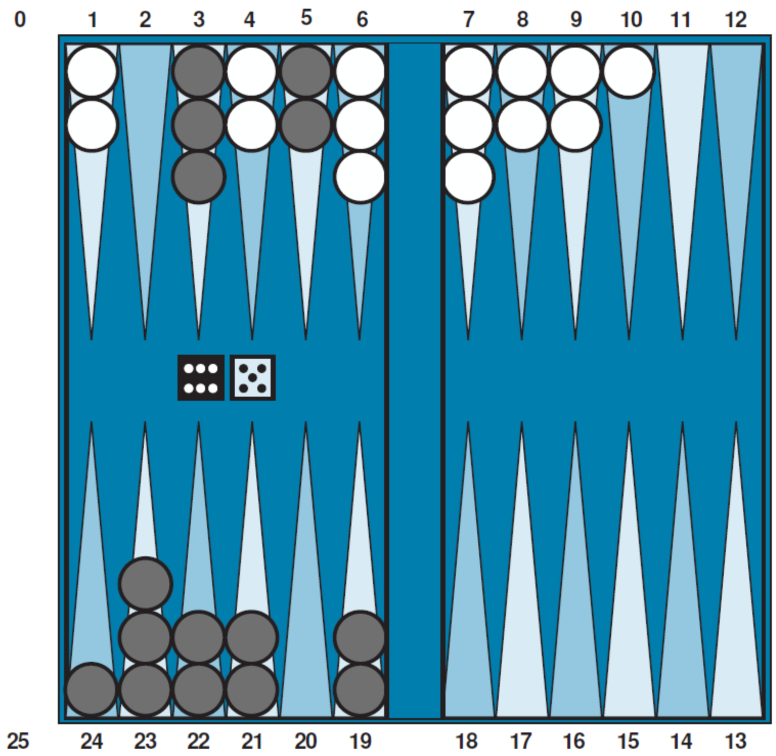
- Include *chance nodes* in addition to MAX and MIN nodes.

Throwing two dice (unordered):

* Figures/images are from the textbook site (or by the instructor).

Stochastic Games

Some games (e.g., backgammon) have randomness due to throwing of dice. They combine both luck and skill.



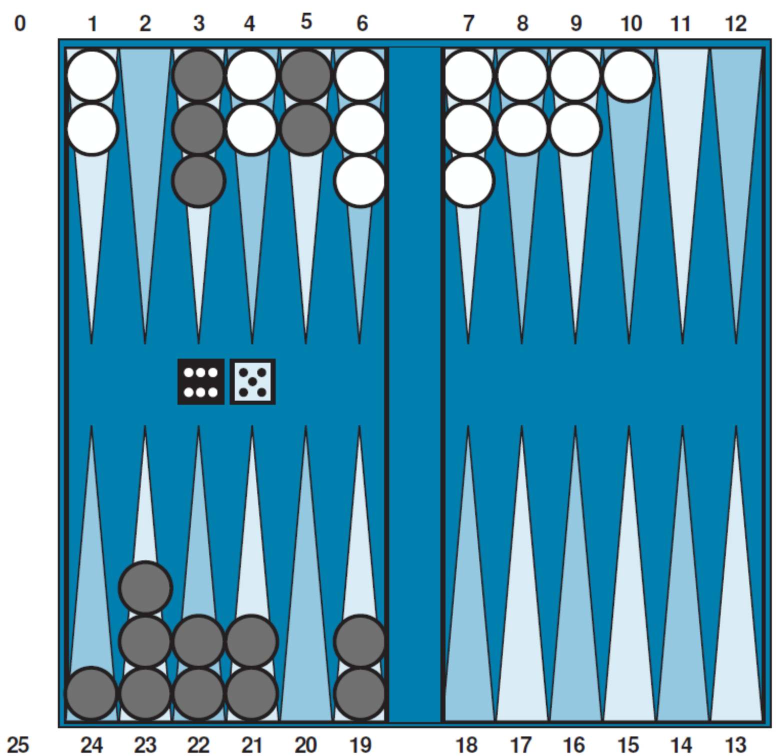
- Include *chance nodes* in addition to MAX and MIN nodes.

Throwing two dice (unordered):

1-1,…,6-6: probability $1/36$ each

Stochastic Games

Some games (e.g., backgammon) have randomness due to throwing of dice. They combine both luck and skill.



- Include *chance nodes* in addition to MAX and MIN nodes.

Throwing two dice (unordered):

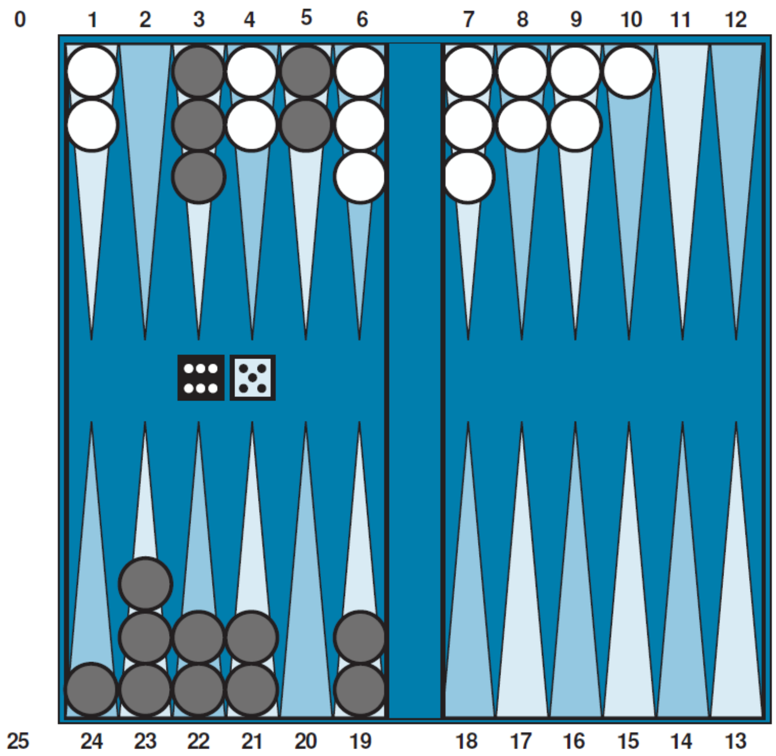
1-1,…,6-6: probability $1/36$ each

1-2,…,1-6, 2-3,…, 5-6: probability $1/18$ each

* Figures/images are from the textbook site (or by the instructor).

Stochastic Games

Some games (e.g., backgammon) have randomness due to throwing of dice. They combine both luck and skill.



- Include *chance nodes* in addition to MAX and MIN nodes.

Throwing two dice (unordered):

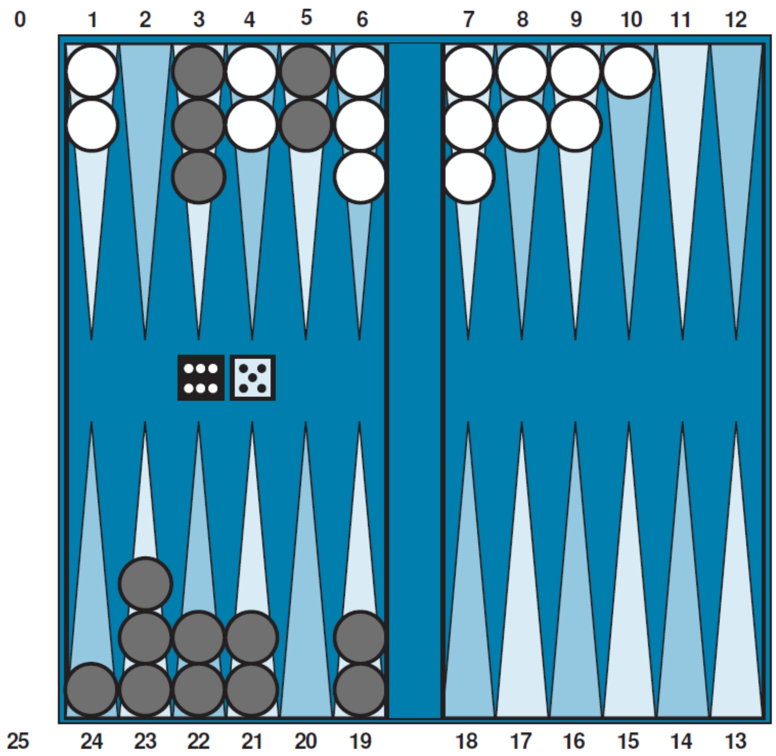
1-1,…,6-6: probability $1/36$ each

$\underbrace{\text{1-2,…,1-6, 2-3,…, 5-6}}_{15}$: probability $1/18$ each

* Figures/images are from the textbook site (or by the instructor).

Stochastic Games

Some games (e.g., backgammon) have randomness due to throwing of dice. They combine both luck and skill.



- Include *chance nodes* in addition to MAX and MIN nodes.

Throwing two dice (unordered):

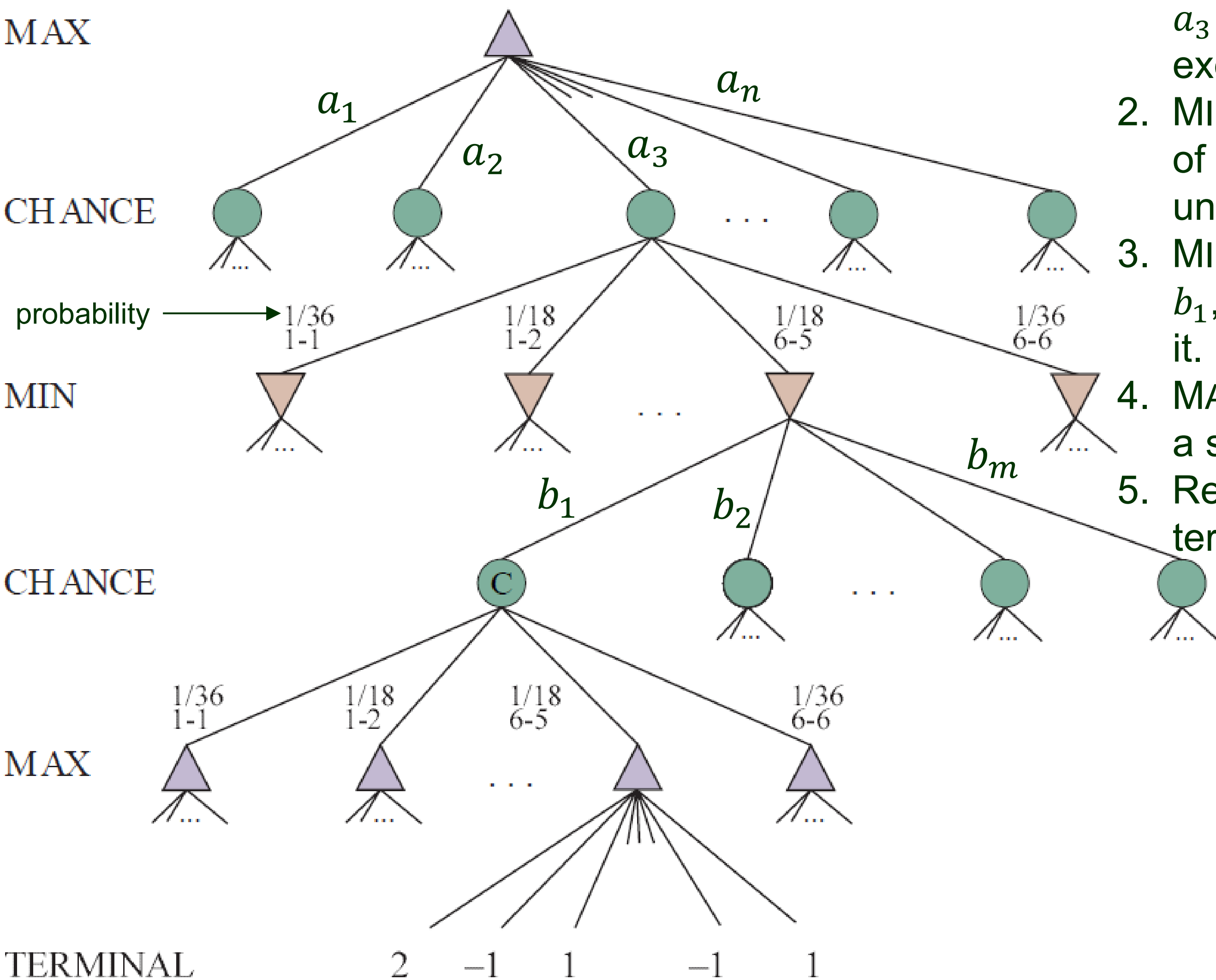
1-1,…,6-6: probability $1/36$ each

$\underbrace{\text{1-2,…,1-6, 2-3,…, 5-6}}_{15}$: probability $1/18$ each

- Calculate expected value (called *expectiminimax* value) of a position.

* Figures/images are from the textbook site (or by the instructor).

Game Tree for a Backgammon Position



1. MAX chooses an action, say, a_3 out of $\{a_1, .., a_n\}$ and executes it.
2. MIN rolls a die and gets a set of legal actions $\{b_1, .., b_m\}$ under some CHANCE node.
3. MIN chooses an action, say, b_1, from the set and executes it.
4. MAX rolls a die and gets a set of legal actions.
5. Repeat steps 1-4 until a terminal state is reached.

Expectiminimax Value

EXPECTIMINIMAX(s) =

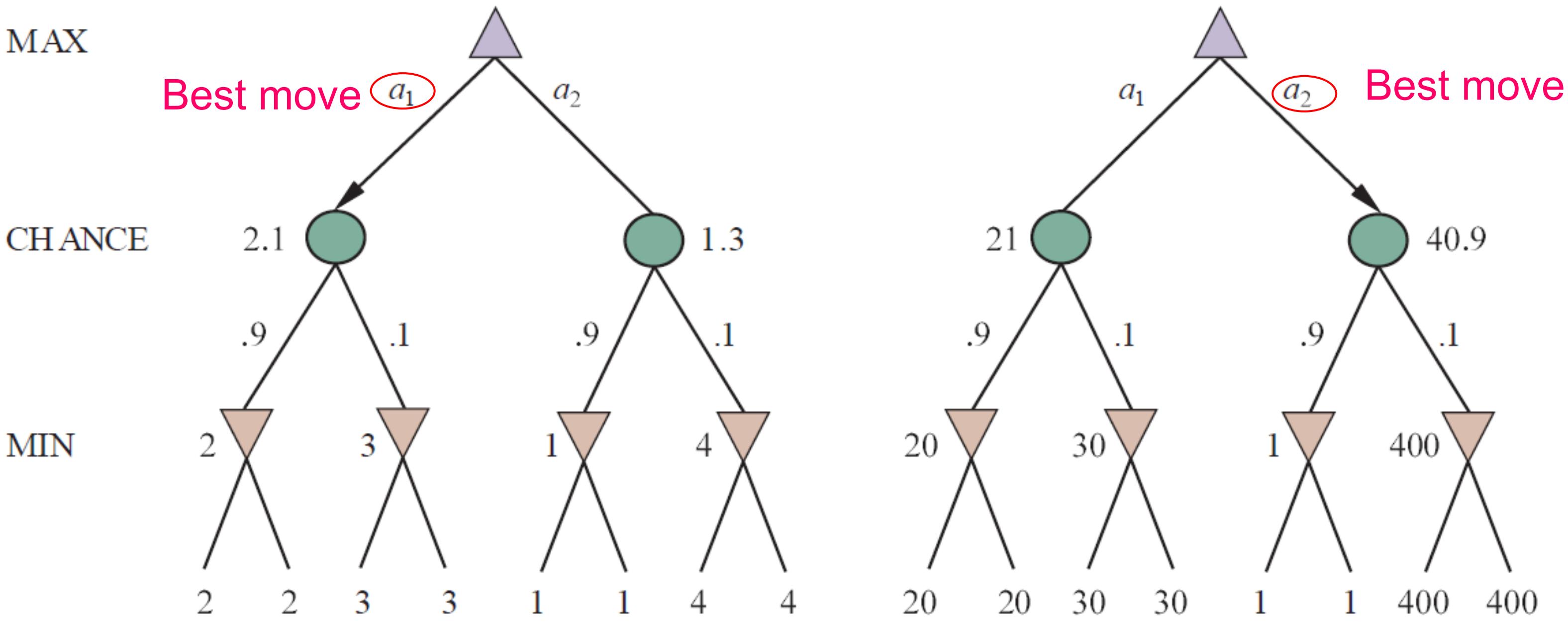
$$\begin{cases} \text{UTILITY}(s, \text{MAX}) & \text{if IS-TERMINAL}(s) \\ \max_a \text{EXPECTIMINIMAX}(\text{RESULT}(s,a)) & \text{if TO-MOVE}(s) = \text{MAX} \\ \min_a \text{EXPECTIMINIMAX}(\text{RESULT}(s,a)) & \text{if TO-MOVE}(s) = \text{MIN} \\ \boxed{\sum_r P(r)\ \text{EXPECTIMINIMAX}(\text{RESULT}(s,r))} & \text{if TO-MOVE}(s) = \text{CHANCE} \end{cases}$$

expected value

one possible dice roll

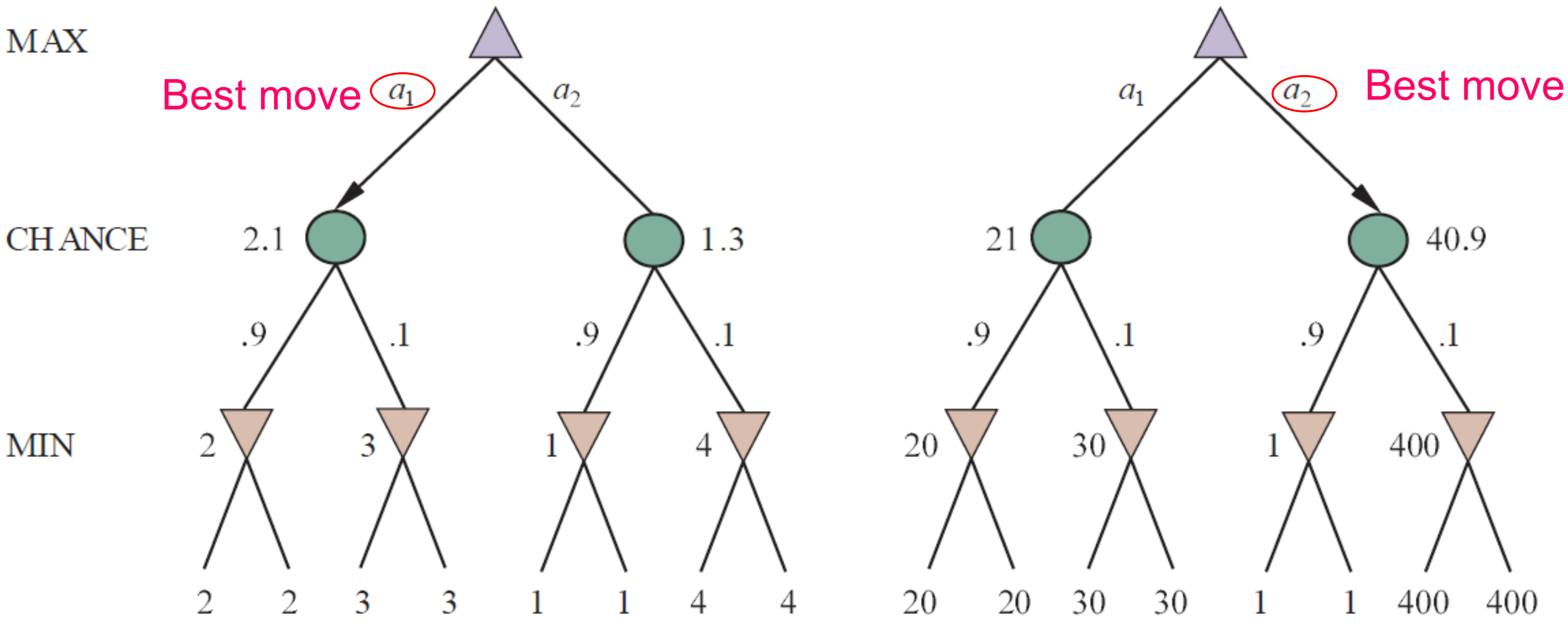
Evaluation Functions

Evaluation functions with the same order of leaf values can yield different move choices at a state.



Evaluation Functions

Evaluation functions with the same order of leaf values can yield different move choices at a state.



Alpha-beta pruning is still applicable if we can bound values on chance nodes.